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Bystanders' responses to offline bullying and cyberbullying: The role of empathy and normative beliefs about aggression

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Cyberbullying often takes place with the virtual presence or knowledge of bystanders. While we have some evidence about the determinants of bystanders' responses to offline bullying, we lack empirical studies concerning the variables that influence bystanders' responses to cyberbullying. The current study examines bystanders' responses to offline bullying and cyberbullying incidents. Two types of responses were captured: support toward the victims and the reinforcement of bullies' actions. Using data from 321 German adolescents (ages 12–18; $M = 14.99$; 44% girls), the association between bystanders' responses and normative beliefs about verbal aggression and cyberaggression, and affective and cognitive empathy, were tested in a path model. Both types of normative beliefs positively predicted the reinforcement of bullies, and normative belief about verbal aggression also predicted support for the victims of offline bullying. Both types of empathy predicted support in offline bullying, but only affective empathy predicted support in cyberbullying. There was no link between affective or cognitive empathy to the reinforcement of bullies. Moreover, bystanders' tendencies to respond supportively to the victim or to reinforce the bully were rather consistent in both cyber- and offline bullying, but there was no link between support and reinforcement. The findings are discussed with regard to implications for prevention and intervention efforts.

Key words: Offline bullying, cyberbullying, bystanders, empathy, normative beliefs about aggression.

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INTRODUCTION

Offline bullying and cyberbullying designate specific types of aggressive behavior that occur among youth. In prior research, both these phenomena have been conceptualized upon the criteria of intention to harm, power imbalance, and repetitiveness (Olweus, 1994); cyberbullying was additionally defined by conducting aggressive attacks via internet or mobile devices (Kowalski, Giumetti, Schroeder & Lattanner, 2014; Tokunaga, 2010). Despite the similarities, offline bullying and cyberbullying are also differentiated by several specific features (see e.g., Cassidy, Faucher & Jackson, 2013; Dooley, Pyżalski & Cross, 2009; Kowalski *et al*., 2014). Unlike offline bullying, in cyberbullying all actors can be mutually distant, invisible, or even anonymous. Cyberbullying has a "24/7 nature" because the attacks can be conducted via technological devices from any place at any time. There is a lack of cues which would provide feedback about the impact on the victim due to the mediated character of attacks via ICT technologies. The harmful materials online can be saved simply and are easily spread and made accessible. Moreover, cyberbullying can potentially involve a wide audience of internet users.

In summary, while offline bullying and cyberbullying share common features (and are, in fact, often also interconnected; Cappadocia, Craig & Pepler, 2013; Modecki, Minchin, Harbaugh, Guerra & Runions, 2014; Waasdorp & Bradshaw, 2015), they present two specific aggressive acts. Previous research focused predominantly on factors connected to the vulnerability or resilience of victims, or the characteristics of the perpetrators (Fanti, Demetriou & Hawa, 2012; Festl & Quandt, 2013; Gradinger, Strohmeier, Schiller, Stefanek & Spiel, 2009; Låftman, Modin & Östberg, 2013; Mishna, Khoury-Kassabri, Gadalla & Daciuk, 2012; Sticca, Ruggieri, Alsaker & Perren 2013). Yet, prior research also emphasized the complexity of offline bullying and cyberbullying as social processes, and the need to also study the reactions of bystanders and cyberbystanders (Cowie, 2014; Pfetsch, 2016; Salmivalli, 2010; 2014). Bystander responses to offline bullying and cyberbullying can crucially influence both bullying processes: they can help to stop the cyber/bullying; help the victim to cope with the incident; or, on the other hand, increase the severity of harm by joining the bullies (Hawkins, Pepler & Craig, 2001; Salmivalli, 2010).

Prior research on offline bullying (or "school bullying" or "traditional bullying") identified several possible ways in which bystanders reacted to ongoing offline bullying. For example, Salmivalli, Lagerspetz, Björkqvist *et al*. (1996) distinguished four participant roles of bystanders: assistants of the bully; reinforcers of the bully; defenders of the victim; and outsiders. Generally, bystanders' responses can be categorized into support for the victim, reinforcement of the bully, and passive response. The supportive responses may include support for the victim in the form of emotional support or advice for how to cope with the incident, confrontation of the bully (physical or verbal, online or offline), or seeking help on the victim's behalf. Reinforcement of the bully may take the form of joining in (e.g., joining in the verbal assaults in offline bullying or adding hurtful comments on the social networking sites in cyberbullying) or support for the bully (e.g., cheering them up or encouraging them by adding "likes" to the online attacks). Passivity may have the form of passive witnessing or withdrawing from the incident.

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In this study, we focus on the supportive and reinforcing responses to offline bullying and cyberbullying. Despite the basic commonalities between these two types of reactions in both processes, it is crucial to also understand the differences which should be considered in relation to offline bullying and cyberbullying. Unlike bystanders in offline bullying, cyberbystanders do not have to be directly present during the attacks. They do not have to see the impact on the victim and, therefore, it might be difficult for them to assess the severity of the attacks. They themselves might not be seen by any other actors, and may be physically distant from them (Macháčková *et al*., 2013; Pfetsch, 2016). Moreover, both types of bullying can be assessed differently. For example, Sticca and Perren (2013) found that cyberbullying might be considered more severe than offline bullying, although they also stressed that the assessment of severity depends on the specific character of the attacks, specifically publicity and anonymity. Considering the differences between offline bullying and cyberbullying, we investigate the role of two factors which have been linked to bystander behavior: normative beliefs about aggression and affective and cognitive empathy. While both of these individual characteristics have been studied in connection with bystander responses to offline bullying, little is known about their links to the responses of cyberbystanders.

NORMATIVE BELIEFS ABOUT AGGRESSION

Normative beliefs are cognitive self-regulatory mechanisms which guide the assessment for what is appropriate or acceptable behavior (Huesmann & Guerra, 1997; Guerra, Huesmann & Hanish, 1995). Since normative beliefs stipulate an (in) appropriate reaction, they are one of the factors which affect responses. It is important to note that normative beliefs are connected to multiple types of responses and situations, and they can be both general and situation-specific (Huesmann & Guerra, 1997). In our study, we focus on normative beliefs about aggression, which have been connected to aggressive behavior among children and youth. Previous studies consistently reported that normative beliefs were associated with offline bullying (Burton, Florell & Wygant, 2013; Cook, Williams, Gendron *et al*., 2011; Guerra, Kim & Sadek, 2010; Guerra, Williams & Sadek, 2011; Kikas, Peets, Tropp & Hinn, 2009; Vernberg & Jacobs, 1999; Werner & Nixon, 2005) and cyberbullying (Ang, Tan & Mansor, 2011; Nicol & Fleming, 2010; Williams & Guerra, 2007; Wright & Li, 2013). A meta-analytic study conducted by Cook *et al*. (2010) also pointed out that other-related cognitions, including normative beliefs about aggression, was one of the strongest individual predictors of offline bullying behaviors. Yet, prior research also uncovered that different types of normative beliefs are specifically linked to certain types of aggressive behavior. Werner and Nixon (2005) found unique association beliefs about relational aggression and relational aggressive behavior, and beliefs about physical aggression and this type of behavior. Wright and Li (2013) found a link between different types of normative beliefs and their corresponding online and offline bullying behavior. Moreover, with regard to studies on cyberbullying perpetration, these authors stressed the need to differentiate the beliefs about face-to-face aggression and cyberaggression because they are distinctively linked to cyberbullying perpetration.

Thus, there is robust existing knowledge about the links between the normative beliefs about aggression and aggressive behaviors. Nevertheless, there is a lack of knowledge which would help the understanding of how normative beliefs connect bystander responses to offline bullying and cyberbullying. Just a handful of studies have examined the role of normative beliefs, overall attitudes, or others-oriented cognitions among bystanders. Salmivalli and Voeten (2004) found that children with anti-bullying attitudes more often defended the victim or stayed passive, and less likely joined the bully in offline bullying. These authors also measured the group norms (indicated by the expected standards of behavior in a group, specifically the assessment of and responding to offline bullying) and found that anti- and pro-bullying group norms also helped to explain the bystander responses. In cyberbullying, Bastiaensens, Pabian, Vandebosch *et al*. (2015) found that joining the bullies (via social pressure) was indirectly connected to the bystanders' perception of friends approving the cyberbullying. Sandstrom, Makover and Bartini (2013) showed that children who perceived that their peers held less prosocial attitudes were less likely to report being defenders and more likely to join offline bullies, but no link was found to passive behavior.

Nevertheless, it should be pointed out that the group norms do not have to be linked with individual normative beliefs (and neither with aggressive behavior; Henry, Guerra, Huesmann *et al*., 2000). Therefore, in our study we focus specifically on individual normative beliefs and their association with bystander responses. Adapting the measurement of normative beliefs from Krahé and Möller (2004), we examine the role of two types of normative beliefs and their appropriateness for verbal aggression and cyberaggression. Considering previous findings, we would expect that both types of normative beliefs will be positively linked with bystanders' reinforcing responses to offline bullies and cyberbullies, and negatively with support toward the victim.

AFFECTIVE AND COGNITIVE EMPATHY

Empathy is the ability to understand and feel the emotions of another person (Batson, 2009). Empathy is a multifaceted phenomenon, including cognitive empathy as the ability to comprehend the emotional situation and take the perspective of a person, and affective empathy as the ability to vicariously experience the emotions of another person (Davis, 1994; Hoffman, 2001). Cognitive empathy leads to higher affective empathy (Batson, Lishner, Carpenter *et al*., 2003) and both dimensions have been found to be negatively connected to aggression (Miller & Eisenberg, 1988), offline bullying (Mitsopoulou & Giovazolias, 2015) and cyberbullying (Del Rey *et al*., 2015). Conversely, they are positively connected to prosocial behavior (Eisenberg & Miller, 1987) and defending behavior in offline bullying (Caravita, Di Blasio & Salmivalli, 2009; Espelage, Mebane & Adams, 2004; Gini, Albiero, Benelli & Altoè, 2007; Nickerson, Mele & Princiotta, 2008; Pöyhönen, Juvonen & Salmivalli, 2010; Van Noorden, Haselager, Cillessen & Bukowski, 2014). In contrast, pro-bullies (assistants, reinforcers) report lower levels of affective and cognitive empathy

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(Gini *et al*., 2007), and outsiders report lower affective empathy (Nickerson *et al*., 2008).

Concerning bystanders in cyberbullying, empathy seems to be one possible protective factor against negative online behavior (such as cyberbullying or passive bystanding) and increases the probability of prosocial online behavior (such as actively supporting the victim) (Pfetsch, 2016). For example, experimental data shows that actively taking the perspective of the cybervictim (cognitive empathy) and feeling the emotions of the harmed person (affective empathy) reduces reinforcement for the cyberbully (i.e., forwarding a degrading photo; Barlińska, Szuster & Winiewski, 2013; 2015). Data from real-life experiences in children and adolescents show that affective empathy (or empathic concern) predicts positively the helping of the victim and negatively the joining in the cyberbullying, as well as doing nothing despite noticing the cyberbullying (Van Cleemput, Vandebosch & Pabian, 2014). Further, the likelihood of actively supporting the cybervictim was positively predicted by the immediate empathic response, even controlling for prosocial behavior tendencies, self-esteem, or fear of intervention (Macháčková *et al*., 2014); and Barhight, Hubbard, and Hyde (2012) found link between emotional response to witnessed bullying and intervention in bullying incidents on behalf of the victim. Moreover, trait affective and cognitive empathy is related to less cyberbullying behavior. In a sample of 979 adolescents, different forms of empathy (affective and cognitive offline empathy in offline settings, affective and cognitive media empathy in online settings) were assessed via self and peer reports (Pfetsch, Müller & Ittel, 2014). Cyberbullies report lower affective offline empathy, affective media empathy, and cognitive media empathy than non-involved students, while their peers report the bullies to have less cognitive offline empathy than non-involved students (Pfetsch *et al*., 2014). In summary, cognitive and affective empathy in bystanders of cyberbullying is hypothesized to be linked to lower reinforcement of the cyberbully and higher support for the victim of cyberbullying.

GENDER AND AGE DIFFERENCES

It has been shown that the bystander responses to offline bullying and cyberbullying can also be associated with gender and age, although the findings are not consistent. Several authors found that support toward the victim was more common among younger children compared to older ones (Menesini, Eslea, Smith *et al*., 1997; O'Connell, Pepler & Craig, 1999; Pozzoli & Gini, 2010; Rigby & Slee, 1991; Thornberg & Jungert, 2013; Trach, Hymel, Waterhouse & Neale, 2010; Van Cleemput *et al*., 2014), although others found opposite pattern (Olenik-Shemesh, Heiman & Eden, 2015) or no substantial association with age (Macháčková *et al*., 2013; Quirk & Campbell, 2014). Some studies found supportive reactions more prevalent among girls (Oh & Hazler, 2009; Olenik-Shemesh *et al*., 2015; Salmivalli *et al*., 1996; Thornberg & Jungert, 2013; Trach *et al*., 2010) and reinforcement of the bully among boys (Quirk & Campbell, 2014); others found no differences (Hawkins *et al*., 2001; Macháčková *et al*., 2013; Menesini *et al*., 1997; Van Cleemput *et al*., 2014). These contradictory findings probably result from different conceptualizations of the key concepts, that is, the definition of (cyber)bullying, the type of responses measured (e.g., active responses as emotional support or more assertive confrontation of the bully); sampling (especially the with regard to the age); and accounting for other explanatory variables. However, considering the possible effect of both gender and age on (cyber)bystander responses, we include these as control variables in our study.

THE AIM OF THIS STUDY

In this article, we focus on supportive and reinforcing responses of adolescent bystanders in offline bullying and cyberbullying. Specifically, our goal is to investigate the role of affective and cognitive empathy, and normative beliefs about verbal aggression and cyberaggression Although there is a robust body of knowledge linking both normative beliefs and empathy to aggressive behaviors, little is known about their associations with bystanders' responses in offline bullying and especially in cyberbullying. Moreover, prior research pointed out possible differences between the different types of aggression related to specific types of both normative beliefs (Wright & Li, 2013) and empathy (Caravita *et al*., 2009; Yeo, Ang, Loh, Fu & Karre, 2011; Van Noorden *et al*., 2014). However, to the best of our knowledge, no study has investigated the link between bystander responses with specific types of normative beliefs (i.e., beliefs about verbal aggression and cyberaggression), and just a handful of studies studied cognitive, as well as affective, empathy (Barlińska, Szuster & Winiewski, 2013; 2015; Pfetsch *et al*., 2014). Therefore, controlling for gender and age, we aim to examine these links. We focus our study on a adolescents because cyberbullying is relatively prevalent in this age group compared to younger ages (Barlett & Coyne, 2014), hence the bystanders at this age may be of special interest. The results of our investigation should help to deepen our knowledge about the role of specific cognitive and affective characteristics in bystander responses, and the differences connected to the specific types of bullying incidents.

METHODS

Procedure

Data were collected from the end of 2013 to the beginning of 2014 in 13 classes of four secondary schools (two gymnasium [advanced academic schools], two integrierte sekundarschule [intermediate academic schools]) in a large city in Germany. Parents and students were informed about the study concerning the theme "bystanders of cyberbullying" and asked for active consent. Trained test administrators implemented the paper-and-pencil questionnaires in the classrooms during a regular school lesson. Completion of the questionnaires took about 25–45 minutes.

Participants

A total of $N = 321$ students aged 12–18 ($M = 14.99$, $SD = 1.64$), 44% girls, participated in the survey, 50% gymnasium students and 50% integrierte sekundarschule students. Ten percent indicated that they were not born in Germany. Most participants reported that they have a computer in their bedroom (86%) and that they have internet access (96%).

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Measures

Offline bullying and cyberbullying were introduced in the questionnaire via behavioral items including the term"bullying" (or "mobbing" in German) without a general definition. Responding to offline bullying and cyberbullying was measured with a German scale adapted from Steffgen, Happ and Pfetsch (2013): *supportive responses to victim of offline bullying* (three items, $M = 0.77$, $SD = 0.86$, $\alpha = 0.75$, for example, "I tried to stop bullying in my class."); *supportive responses to victim of cyberbullying* (three items, $M = 0.53$, $SD = 0.72$, $\alpha = 0.78$, for example, "I tried to stop bullying via Facebook and/or YouTube."); *reinforcement of the bully in offline bullying* (three items, $M = 0.37$, $SD = 0.68$, $\alpha = 0.80$ for example, "I joined in humiliating some classmates in the schoolyard."); and *reinforcement of the bully in cyberbullying* (three items, $M = 0.27$, $SD = 0.51$, $\alpha = 0.72$, for example, "I joined in humiliating some classmates via Facebook'). Participants indicated how often in their school career they showed these different responses on a five-point-Likert scale with the options: *never* (0), *one or two times in a school year* (1), *once a month* (2), *several times a week* (3), *several times a day* (4).

Empathy was measured by a back and forth translated and shortened version of the Basic Empathy Scale (Jolliffe & Farrington, 2006), including only positively phrased items in order to reduce cognitive load for the younger participants. Two dimensions were measured: *affective empathy* (four items, $M = 2.41$, $SD = 0.77$, $\alpha = 0.68$, for example, "I get caught up in other people's feelings easily.") and *cognitive empathy* (four items, $M = 2.90$, $SD = 0.77$, $\alpha = 0.76$, for example, "I can often understand how people are feeling even before they tell me.") on a four-point-Likert scale from *does not apply at all* (1) to *applies completely* (4).

Normative beliefs about aggression were measured with an adapted German scale by Krahé and Möller (2004). Participants read the following scenario: "Imagine, you are really mad about a classmate, because he or she has treated you really mean and unfair this morning. All your classmates have seen how he or she has made fun of you. In the afternoon, you accidentally meet him or her outside of school. This time you both are alone. You directly get into a heated argument and he or she provokes you again." Afterwards they indicated how much they approve of several behaviors for how to react on a four-point-Likert scale from *does not apply at all* (1) to *applies completely* (4) which resulted in two subscales, *normative beliefs about verbal aggression* (three items, $M = 2.34$, $SD = 0.86$, $\alpha = 0.80$, e.g., "In this situation I think it is ok … to yell at him or her."), and *normative beliefs about cyberaggression* (four items, $M = 1.42$, $SD = .61$, $\alpha = 0.82$, e.g., "… to degrade him or her later on the internet').

RESULTS

First, we conducted a bivariate analysis in SPSS 20 (IBM, Armonk, NY) (Table 1). Support to the victim of offline bullying and cyberbullying, as well as reinforcement of the offline and cyberbully, were strongly correlated. There was also a weaker but significant correlation ($r = 0.128$, $p < 0.05$) between supportive and reinforcing response in cyberbullying, but not in offline bullying. Both types of empathy were mutually correlated and linked to both supportive responses, but not to reinforcing ones. Normative beliefs about aggression were also correlated and linked with reinforcing responses; beliefs about verbal aggression were additionally associated with cognitive empathy and support to the victim of offline bullying; beliefs about cyberaggression were negatively correlated with cognitive empathy.

To analyse the relationship between responding to (cyber) bullying, empathy, and normative beliefs about aggression, we examined a path model with observed variables using MPlus 6.1. (Muthén & Muthén, Los Angeles, CA). Using MLR as an estimator, we predicted supportive and reinforcing responses for both types of normative beliefs about aggression and both types of empathy; gender and age were also included as predictors to control for their potential effect. In the model (Fig. 1), only the covariance between gender and age (which was non-significant also in the bivariate analysis) was constrained to zero.

This estimated model had a good fit: RMSEA = 0.042, CFI = 0.998, TLI = 0.946, chi square = 1.553, $p = 0.212$. In the model, affective and cognitive empathy predicted supportive responses to the victim of offline bullying, but only affective empathy predicted supportive response to the victim of cyberbullying. There was no link to reinforcing responses. Both types of normative beliefs predicted reinforcing responses, yet not the supportive ones. Moreover, aggressive beliefs about verbal norms also predicted support for the victim of offline bullying. There was a positive association between supportive responses in offline bullying and cyberbullying, and between reinforcing responses in offline bullying and cyberbullying, but there was no link between the supportive and reinforcing responses in either type of bullying.

DISCUSSION

Our study examined the role of empathy and normative beliefs about aggression in bystander responses to offline bullying and cyberbullying. We found that bystanders' tendencies to respond supportively to the victim or reinforcing the bully are rather consistent in cases of both cyberbullying and offline bullying, but there was no link between overall support and reinforcement. The effects of normative beliefs and empathy differed according to the

Table 1. *Correlations among examined variables*

	1	2	3	4	5	6	7	8	9
1. Support for victim of offline bullying	1								
2. Support for victim of cyberbullying	0.499**	1							
3. Reinforcement of bully in offline bullying	0.070	0.078	1						
4. Reinforcement of bully in cyberbullying	0.010	0.128*	0.570	1					
5. Cognitive empathy	0.341**	0.207**	−0.086	−0.078	1				
6. Affective empathy	0.329**	0.263**	−0.089	−0.044	0.597**	1			
7. Normative beliefs about verbal aggression	0.218**	0.054	0.219**	0.188**	0.236**	0.089	1		
8. Normative beliefs about cyberaggression	0.040	0.066	0.407**	0.469**	−0.128*	−0.056	0.158**	1	
9. Gender	0.176**	0.150**	−0.075	−0.071	0.183**	0.402**	−0.075	−0.087	1
10. Age	0.005	0.034	0.05	0.077	0.264**	0.219**	0.145**	−0.009	−0.070

Note: $^{**}p < 0.01$, $^{*}p < 0.05$.

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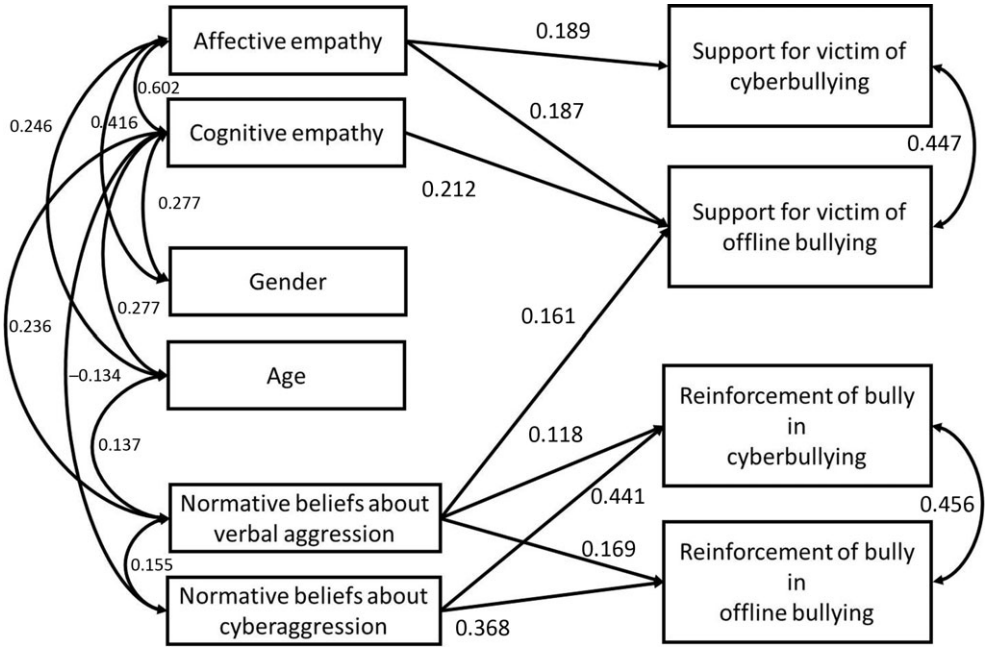


Fig. 1. Model results.
Note: Only significant ($p < 0.05$) paths and associations are displayed.

type of behavior (support vs. reinforcement), but also according to the character of the bullying (cyber vs. offline).

Concerning the overall responses, we found that youth responded similarly to cyberbullying and offline bullying. Adolescents who responded supportively to the victim in offline bullying tended to also do so in cyberbullying, a pattern which also applied for the reinforcement of the bully. Prior research showed that the experiences with offline bullying and cyberbullying were linked among victims and perpetrators (Waasdorp & Bradshaw, 2015), and, according to our findings, this might apply also for the experiences of (cyber)bystanders. However, the overall responses had no linear association, although one could expect that bystanders who respond supportively to the victim would less often reinforce the bully. However, we should stress that the responses were measured with relation to the time frame "school career.", which might include different types of incidents (and probably actors). Thus, although a bystander might have reacted supportively in certain incidents of cyber/bullying, they might react differently in others. This finding could be explained by the diversity of incidents captured in our measurements, since bystander responses differ in relation to different contextual factors. These were outlined in a conceptual framework for cyberbystanders and include: (1) virtuality – the online and offline bystanding to instances of cyberbullying; (2) proximity – the temporal, technical, and psychological closeness of cyberbystanders to the involved persons (cyberbully, cybervictim, and other possible cyberbystanders); and (3) existing social relationship to cybervictims, cyberbullies, and/or other cyberbystanders (Pfetsch, 2016). For example, the type of relationship between victims and bullies (friendship, negative or no relationship) influences the behavior of bystanders to offline bullying and cyberbullying (Desmet, Bastiaensens, Van Cleemput *et al*., 2012; Lodge & Frydenberg, 2005; Macháčková *et al*., 2013; Oh & Hazler, 2009).

Moreover, we uncovered several associations between the specific types of responses and individual characteristics of the bystanders. Normative beliefs have been positively connected with reinforcing the bully, which is in accordance with previous findings on links with aggression (Burton *et al*., 2013; Cook *et al*., 2010; Guerra *et al*., 2011; Werner & Nixon, 2005). So, the more participants approved of aggressive responses in relation to a provocation (high normative beliefs), the more they reported reinforcement as bystanders of cyberbullying. However, with one exception, normative beliefs were not connected to the support for the victim. In other words, while participants high in normative beliefs did not display less supportive responses, even children low on these beliefs did not intervene when someone else was (cyber) bullied. Thus, while normative beliefs might consistently increase reinforcement of the bully, their effect on support for the victims is diverse. It is possible that, although children might not see aggressive attacks as appropriate, they do not intervene because of the additional contextual factors mentioned above. These findings might also reflect a general discrepancy between attitudes or norms, and the actual bystander behavior (Salmivalli & Voeten, 2004).

However, there was one exception to this pattern. We found a link between normative beliefs about verbal aggression and responses to offline bullying – yet, contrary to our expectations, this link was positive. We also found a positive association between these beliefs and cognitive empathy. Since we measured normative beliefs as the appropriateness of aggressive response upon provocation, it is possible that in bystanders such responsiveness can take the role of confronting a bully (and overall assertive defence for the victim) upon the understanding of bullying as inappropriate. In other words, the bystanders who believed that it is appropriate to react by verbal aggression could be more prone to assertively/aggressively defend the victim in front of the bully. However, no such link was found with normative beliefs about cyberaggression, and neither with

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supportive responses in cyberbullying. Therefore, we would support the conclusion of previous research that it is important to distinguish among specific types of normative beliefs, since they may differ in their effect on specific types of youth behavior (Werner & Nixon, 2005; Wright & Li, 2013). Further, in future studies we would suggest to capture (cyber)bystanders' assessment of the situation, including the attribution of the cause of the incident, that is, whether they thought that the aggressor was behaving inappropriately, or whether his or hers actions were perceived as an appropriate response and bystanders actually blamed the victim. These attributions could moderate the association between normative beliefs and bystanders' responses, and such investigation could overall help us to further understand under which circumstances bystanders provide support to victim and under which circumstances they reinforce the bullies.

The need for more specific differentiation also concerns empathy. Overall, empathy was associated only with support for the victim, with no consistent relation to reinforcement of the bully. Thus, while increased empathy is connected with supportive responses (as in prior research; Gini *et al*., 2007; Pöyhönen et al, 2010; Van Cleemput *et al*., 2014), it does not ensure that children would not reinforce the bully. Moreover, while affective empathy predicted supportive responses in both types of incidents, cognitive empathy was associated only with supportive responses to offline bullying. In prior research, it has been shown that the immediate affective empathic or emotional response plays a significant role in providing support to victims (Barhight *et al*., 2012; Macháčková *et al*., 2013). Thus, while cognitive empathy could facilitate immediate affective empathy, affective empathy seems to be a relevant determining factor for the support of the victim. However, in cyberbullying it is often more difficult than in offline bullying to assess the severity of the incident (e.g., due to the distance and the invisibility of actors; Pfetsch, 2016). Therefore, possibly even children higher on cognitive empathy might incorrectly assess the severity of the incident as being low, leading to low immediate affective empathy and low supportive responses in turn. The finding that neither affective nor cognitive empathy was related to reinforcement of bullies in (cyber)bullying stands in contrast to previous research (Barlińska *et al*., 2013; 2015; Van Cleemput *et al*., 2014). However, the findings may be due to different operationalization, like a different trait empathy scale (Van Cleemput *et al*., 2014) or a state measure for empathy (Barlińska *et al*., 2013; 2015). Further, the influence of normative beliefs seems to be so strong that empathy did not play a significant role in bystanders' reinforcing responses: Normative influences might override the influence of empathy in the bystander situation. These possible explanations should be followed in future research.

The demographics of age and gender had no effect on our measured outcomes, although they were connected to our predictors and outcomes on a bivariate level. Yet, considering that we found no association with demographics in the path model, we would suggest that the cognitive and affective factors are more important factors for understanding bystanders' responses.

The interpretation of our findings should also acknowledge that our study has several limitations. The cross-sectional design limits the interpretation in terms of causality. Experimental and longitudinal studies should corroborate our findings concerning the reciprocal influences of the variables in controlled circumstances and over time. We generally differentiated between support for the victim and reinforcement of the bully, but there can also be more nuanced differences regarding the specific subtypes of these behaviors (as described in the introduction). Further, we examined the responses based on self-reported data, which could be partly biased due to social desirability. Finally, the inclusion of contextual factors, like peer groups,' parents,' or teachers' influence, which could moderate the effect of individual characteristics, would further deepen our understanding of the bystanders' response to (cyber)bullying.

Despite these limitations, we believe that our study sheds more light on the question of how adolescents respond to offline bullying and cyberbullying. Empathic skills seem especially relevant for supportive responses toward the victim, whereas normative beliefs about aggression more strongly influence the reinforcement of the (cyber)bully. Considering our findings, we can also formulate several implications for the area of prevention and intervention. In our study, empathy has an important role for helping the victim, and in line with prior studies (Ang & Goh, 2010; Barlińska *et al*., 2015; Steffgen, König, Pfetsch & Melzer, 2011), we would encourage efforts to increase both types of empathy, since this could increase supportive bystander response. Practical exercises to put oneself into the shoes of a communication partner in computer-mediated communication or to formulate the thoughts, emotions, and wishes of (cyber) victims might help to reach this goal. Nevertheless, this does not ensure that children will not reinforce the (cyber)bullies, especially if they believe that aggressive responses are appropriate. Thus, professionals working with children and adolescents should also help them understand that a supportive response for (cyber)victims is appropriate – a belief which might have been reflected in the normative beliefs in verbal aggression in our study. Nevertheless, it is vital to help children (especially those with high normative beliefs about aggression) understand that support should rather take the non-aggressive form, such as non-aggressive verbal confrontation with the bully. Youth could also be informed properly about the hurtful nature of cyberbullying, since they might struggle with accurate assessments of the incidents, even if they otherwise show deeper understanding of others' harm. Establishing an anti-bullying and pro-intervention norm for bystanders of offline bullying as well as cyberbullying in group contexts like school classes, sport teams, or youth groups could help in reducing harmful experiences for children and adolescents. Peers themselves may be a valuable resource for this aim, and targeting peer norms can be an effective way to affect personal normative beliefs of adolescents.

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REFERENCES

Ang, R. P. & Goh, D. H. (2010). Cyberbullying among adolescents: The role of affective and cognitive empathy, and gender. *Child Psychiatry & Human Development*, *41*, 387–397.

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Ang, R. P., Tan, K. A. & Mansor, A. T. (2011). Normative beliefs about aggression as a mediator of narcissistic exploitativeness and cyberbullying. *Journal of Interpersonal Violence*, *26*, 2619–2634.

Barhight, L. R., Hubbard, J. A. & Hyde, C. T. (2012). Children's physiological and emotional reactions to witnessing bullying predict bystander intervention. *Child Development*, *84*, 375–390.

Barlett, C. & Coyne, S. M. (2014). A meta-analysis of sex differences in cyber-bullying behavior: The moderating role of age. *Aggressive Behavior*, *40*, 474–488.

Barlińska, J., Szuster, A. & Winiewski, M. (2013). Cyberbullying among adolescent bystanders: Role of the communication medium, form of violence, and empathy. *Journal of Community & Applied Social Psychology*, *23*, 37–51.

Barlińska, J., Szuster, A. & Winiewski, M. (2015). The role of short- and long-term cognitive empathy activation in preventing cyberbystander reinforcing cyberbullying behavior. *Cyberpsychology, Behavior, and Social Networking*, *18*, 241–244.

Bastiaensens, S., Pabian, S., Vandebosch, H., Poels, K., Van Cleemput, K., DeSmet, A. & De Bourdeaudhuij, I. (2015). From normative influence to social pressure: How relevant others affect whether bystanders join in cyberbullying. *Social Development*, *25*, 193–211.

Batson, C. D. (2009). These things called empathy: Eight related but distinct phenomena. In J. Decety & W. Ickes (Eds.), *The social neuroscience of empathy* (pp. 3–15). Cambridge, MA: MIT Press.

Batson, C. D., Lishner, D. A., Carpenter, A., Dulin, L., Harjusola-Webb, S., Stocks, E. L. *et al*. (2003). "As you would have them do unto you": Does imagining yourself in the other's place stimulate moral action? *Personality and Social Psychology Bulletin*, *29*, 1190–1201.

Burton, K. A., Florell, D. & Wygant, D. B. (2013). The role of peer attachment and normative beliefs about aggression on traditional bullying and cyberbullying. *Psychology in the Schools*, *50*, 103–115.

Cappadocia, M. C., Craig, W. M. & Pepler, D. (2013). Cyberbullying prevalence, stability, and risk factors during adolescence. *Canadian Journal of School Psychology*, *28*, 171–192.

Caravita, S. S., Di Blasio, P. & Salmivalli, C. (2009). Unique and interactive effects of empathy and social status on involvement in bullying. *Social Development*, *18*, 140–163.

Cassidy, W., Faucher, C. & Jackson, M. (2013). Cyberbullying among youth: A comprehensive review of current international research and its implications and application to policy and practice. *School Psychology International*, *34*, 575–612.

Cook, C. R., Williams, K. R., Guerra, N. G., Kim, T. E. & Sadek, S. (2010). Predictors of bullying and victimization in childhood and adolescence: A meta-analytic investigation. *School Psychology Quarterly*, *25*, 65–83.

Cowie, H. (2014). Understanding the role of bystanders and peer support in school bullying. *The International Journal of Emotional Education*, *6*, 26–32.

Davis, M. H. (1994). *Empathy: A social psychological approach*. Dubugue, IA: Westview Press.

Del Rey, R., Lazuras, L., Casas, J. A., Barkoukis, V., Ortega-Ruiz, R. & Tsorbatzoudis, H. (2015). Does empathy predict (cyber) bullying perpetration, and how do age, gender and nationality affect this relationship? *Learning and Individual Differences*, *45*, 275–281.

Desmet, A., Bastiaensens, S., Van Cleemput, K., Poels, K., Vandebosch, H. & De Bourdeaudhuij, I. (2012). Mobilizing bystanders of cyberbullying: An exploratory study into behavioural determinants of defending the victim. In B. K. Wiederhold & G. Riva (Eds.), *Annual review of cybertherapy and telemedicine* (pp. 58–63). Amsterdam: IOS Press.

Dooley, J. J., Pyżalski, J. & Cross, D. (2009). Cyberbullying versus face-to-face bullying: A theoretical and conceptual review. *Zeitschrift für Psychologie/Journal of Psychology*, *217*, 182–188.

Eisenberg, N. & Miller, P. A. (1987). The relation of empathy to prosocial and related behaviors. *Psychological bulletin*, *101*, 91–119.

Espelage, D. L., Mebane, S. E. & Adams, R. S. (2004). Empathy, caring, and bullying: Toward an understanding of complex associations. In D.L. Espelage & S.M. Swearer (Eds.), *Bullying in American schools: A social-ecological perspective on prevention and intervention* (pp. 37–61). Mahwah, NJ: Lawrence.

Fanti, K. A., Demetriou, A. G. & Hawa, V. V. (2012). A longitudinal study of cyberbullying: Examining risk and protective factors. *European Journal of Developmental Psychology*, *9*, 168–181.

Festl, R. & Quandt, T. (2013). Social relations and cyberbullying: The influence of individual and structural attributes on victimization and perpetration via the Internet. *Human Communication Research*, *39*, 101–126.

Gendron, B. P., Williams, K. R. & Guerra, N. G. (2011). An analysis of bullying among students within schools: Estimating the effects of individual normative beliefs, self-esteem, and school climate. *Journal of School Violence*, *10*, 150–164.

Gini, G., Albiero, P., Benelli, B. & Altoè, G. (2007). Does empathy predict adolescents' bullying and defending behavior? *Aggressive Behavior*, *33*, 467–476.

Gradinger, P., Strohmeier, D., Schiller, E. M., Stefanek, E. & Spiel, C. (2012). Cyber-victimization and popularity in early adolescence: Stability and predictive associations. *European Journal of Developmental Psychology*, *9*, 228–243.

Guerra, N. G., Huesmann, L. R. & Hanish, L. (1995). The role of normative beliefs in children's social behavior. In N. Eisenberg (Ed.), *Social development* (pp. 140–158). Thousnads Oaks, CA, London, UK, New Dehli, India: Sage.

Guerra, N. G., Williams, K. R. & Sadek, S. (2011). Understanding bullying and victimization during childhood and adolescence: A mixed methods study. *Child Development*, *82*, 295–310.

Hawkins, D. L., Pepler, D. J. & Craig, W. M. (2001). Naturalistic observations of peer interventions in bullying. *Social Development*, *10*, 512–527.

Henry, D., Guerra, N., Huesmann, R., Tolan, P., Van Acker, R. & Eron, L. (2000). Normative influences on aggression in urban elementary school classrooms. *American Journal of Community Psychology*, *28*, 59–81.

Hoffman, M. L. (2001). *Empathy and moral development: Implications for caring and justice*. Cambridge: Cambridge University Press.

Huesmann, L. R. & Guerra, N. G. (1997). Children's normative beliefs about aggression and aggressive behavior. *Journal of personality and social psychology*, *72*, 408–419.

Jolliffe, D. & Farrington, D. P. (2006). Development and validation of the basic empathy scale. *Journal of Adolescence*, *29*, 589–611.

Kikas, E., Peets, K., Tropp, K. & Hinn, M. (2009). Associations between verbal reasoning, normative beliefs about aggression, and different forms of aggression. *Journal of Research on Adolescence*, *19*, 137–149.

Kowalski, R. M., Giumetti, G. W., Schroeder, A. N. & Lattanner, M. R. (2014). Bullying in the digital age: A critical review and meta-analysis of cyberbullying research among youth. *Psychological Bulletin*, *140*, 1073–1137.

Krahé, B. & Möller, I. (2004). Playing violent electronic games, hostile attributional style, and aggression-related norms in German adolescents. *Journal of Adolescence*, *27*, 53–69.

Låftman, S. B., Modin, B. & Östberg, V. (2013). Cyberbullying and subjective health: A large-scale study of students in Stockholm, Sweden. *Children and Youth Services Review*, *35*, 112–119.

Lodge, J. & Frydenberg, E. (2005). The role of peer bystanders in school bullying: Positive steps toward promoting peaceful schools. *Theory into Practice*, *44*, 329–336.

Macháčková, H., Dedkova, L., Sevcikova, A. & Cerna, A. (2013). Bystanders' support of cyberbullied schoolmates. *Journal of Community & Applied Social Psychology*, *23*, 25–36.

Macháčková, H., Dedková, L., Ševcíková, A. & Cerná, A. (2014). *Active and Passive Cyber-Bystanders: What is the Difference?* Paper presented at the Cyberspace conference conference, November, Brno.

Menesini, E., Eslea, M., Smith, P. K., Genta, M. L., Giannetti, E., Fonzi, A. & Costabile, A. (1997). Cross-national comparison of children's attitudes towards bully/victim problems in school. *Aggressive Behavior*, *23*, 245–257.

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14679450, 2016, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sjop.12277 by John Carroll University, Wiley Online Library on [06/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Miller, P. A. & Eisenberg, N. (1988). The relation of empathy to aggressive and externalizing/antisocial behavior. *Psychological Bulletin*, *103*, 324–344.

Mishna, F., Khoury-Kassabri, M., Gadalla, T. & Daciuk, J. (2012). Risk factors for involvement in cyber bullying: Victims, bullies and bully-victims. *Children and Youth Services Review*, *34*, 63–70.

Mitsopoulou, E. & Giovazolias, T. (2015). Personality traits, empathy and bullying behavior: A meta-analytic approach. *Aggression and Violent Behavior*, *21*, 61–72.

Modecki, K. L., Minchin, J., Harbaugh, A. G., Guerra, N. G. & Runions, K. C. (2014). Bullying prevalence across contexts: A meta-analysis measuring cyber and traditional bullying. *Journal of Adolescent Health*, *55*, 602–611.

Nickerson, A. B., Mele, D. & Princiotta, D. (2008). Attachment and empathy as predictors of roles as defenders or outsiders in bullying interactions. *Journal of School Psychology*, *46*, 687–703.

Nicol, A. & Fleming, M. J. (2010). "i h8 u": The influence of normative beliefs and hostile response selection in predicting adolescents' mobile phone aggression – a pilot study. *Journal of School Violence*, *9*, 212–231.

O'Connell, P., Pepler, D. & Craig, W. (1999). Peer involvement in bullying: Insights and challenges for intervention. *Journal of Adolescence*, *22*, 437–452.

Oh, I. & Hazler, R. J. (2009). Contributions of personal and situational factors to bystanders' reactions to school bullying. *School Psychology International*, *30*, 291–310.

Olenik-Shemesh, D., Heiman, T. & Eden, S. (2015). Bystanders' behavior in cyberbullying episodes active and passive patterns in the context of personal-socio-emotional factors. *Journal of Interpersonal Violence*.

Olweus, D. (1994). Bullying at school. Long-term outcomes for the victims and an effective school-based intervention program. In L. R. Huesmann (Ed.), *Aggressive behavior: Current perspectives* (pp. 97–130). New York: Springer US.

Pfetsch, J. (2016). Who is who in cyberbullying? Conceptual and empirical perspectives on bystanders in cyberbullying. In M. F. Wright (Ed.), *A social-ecological approach to cyberbullying* (pp. 121–149). Hauppauge: Nova Publishing.

Pfetsch, J., Müller, C. R. & Ittel, A. (2014). cyberbullying und empathie: Affektive, kognitive und medienbasierte empathie im kontext von cyberbullying im kindes- und jugendalter[Cyberbullying and empathy: Affective, cognitive and media-based empathy in the context of cyberbullying in children and adolescents]. *Diskurs Kindheits- und Jugendforschung*, *9*, 23–37.

Pöyhönen, V., Juvonen, J. & Salmivalli, C. (2010). What does it take to stand up for the victim of bullying? The interplay between personal and social factors. *Merrill-Palmer Quarterly*, *56*, 143–163.

Pozzoli, T. & Gini, G. (2012). Why do bystanders of bullying help or not? A multidimensional model. *The Journal of Early Adolescence*, *33*, 315–340.

Quirk, R. & Campbell, M. (2015). On standby? A comparison of online and offline witnesses to bullying and their bystander behaviour. *Educational Psychology*, *35*, 430–448.

Rigby, K. & Slee, P. T. (1991). Bullying among Australian schoolchildren: Reported behavior and attitudes toward victims. *The Journal of Social Psychology*, *131*, 615–627.

Salmivalli, C. (2010). Bullying and the peer group: A review. *Aggression and Violent Behavior*, *15*, 112–120.

Salmivalli, C. (2014). Participant roles in bullying: How can peer bystanders be utilized in interventions? *Theory Into Practice*, *53*, 286–292.

Salmivalli, C., Lagerspetz, K., Björkqvist, K., Österman, K. & Kaukiainen, A. (1996). Bullying as a group process: Participant roles and their relations to social status within the group. *Aggressive Behavior*, *22*, 1–15.

Salmivalli, C. & Voeten, M. (2004). Connections between attitudes, group norms, and behaviour in bullying situations. *International Journal of Behavioral Development*, *28*, 246–258.

Sandstrom, M., Makover, H. & Bartini, M. (2013). Social context of bullying: Do misperceptions of group norms influence children's responses to witnessed episodes? *Social Influence*, *8*(2–3), 196–215.

Steffgen, G., Happ, C. & Pfetsch, J. (2013). Measuring bystander behavior in bullying and cyberbullying incidents. In *Cyberbullying definition and measurement: State of the art and new challenges. Symposium conducted at the XVI European Conference on Developmental Psychology, September*, Lausanne.

Steffgen, G., König, A., Pfetsch, J. & Melzer, A. (2011). Are cyberbullies less empathic? Adolescents' cyberbullying behavior and empathic responsiveness. *Cyberpsychology, Behavior, and Social Networking*, *14*, 643–648.

Sticca, F. & Perren, S. (2013). Is cyberbullying worse than traditional bullying? Examining the differential roles of medium, publicity, and anonymity for the perceived severity of bullying. *Journal of Youth and Adolescence*, *42*, 739–750.

Sticca, F., Ruggieri, S., Alsaker, F. & Perren, S. (2013). Longitudinal risk factors for cyberbullying in adolescence. *Journal of Community & Applied Social Psychology*, *23*, 52–67.

Thornberg, R. & Jungert, T. (2013). School bullying and the mechanisms of moral disengagement. *Aggressive Behavior*, *40*, 99–108.

Tokunaga, R. S. (2010). Following you home from school: A critical review and synthesis of research on cyberbullying victimization. *Computers in Human Behavior*, *26*, 277–287.

Trach, J., Hymel, S., Waterhouse, T. & Neale, K. (2010). Bystander responses to school bullying: A cross-sectional investigation of grade and sex differences. *Canadian Journal of School Psychology*, *25*, 114–130.

Van Cleemput, K., Vandebosch, H. & Pabian, S. (2014). Personal characteristics and contextual factors that determine "helping", "joining in", and "doing nothing" when witnessing cyberbullying. *Aggressive Behavior*, *40*, 383–396.

Van Noorden, T. H. J., Haselager, G. J. T., Cillessen, A. H. N. & Bukowski, W. M. (2014). Empathy and involvement in bullying in children and adolescents: A systematic review. *Journal of Youth and Adolescence*, *44*, 637–657.

Vernberg, E. & Jacobs, A. (1999). Peer victimization and attitudes about violence during early adolescence. *Journal of Clinical Child Psychology*, *28*, 386–395.

Waasdorp, T. E. & Bradshaw, C. P. (2015). The overlap between cyberbullying and traditional bullying. *Journal of Adolescent Health*, *56*, 483–488.

Werner, N. E. & Nixon, C. L. (2005). Normative beliefs and relational aggression: An investigation of the cognitive bases of adolescent aggressive behavior. *Journal of Youth and Adolescence*, *34*, 229–243.

Williams, K. R. & Guerra, N. G. (2007). Prevalence and predictors of internet bullying. *Journal of Adolescent Health*, *41*, S14–S21.

Wright, M. F. & Li, Y. (2013). Normative beliefs about aggression and cyber aggression among young adults: A longitudinal investigation. *Aggressive Behavior*, *39*, 161–170.

Yeo, L. S., Ang, R. P., Loh, S., Fu, K. J. & Karre, J. K. (2011). The role of affective and cognitive empathy in physical, verbal, and indirect aggression of a Singaporean sample of boys. *The Journal of Psychology*, *145*, 313–330.

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